

THIRUVENKADAPRASATH.S

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14	Implement House price Prediction using appropriate machine learning algorithm
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CSV

SquareFeet,Bedrooms,Age,Price

1500,3,10,300000

2000,4,5,400000

1100,2,20,220000

2500,4,2,500000

1800,3,15,340000

1300,2,8,260000

2200,4,12,430000

3000,5,1,600000

PROGRAM

```
import numpy as np
```

```
import pandas as pd
```

```
class RidgeRegressionScratch:
```

```
    def __init__(self, alpha=1.0):
```

```
        self.alpha = alpha
```

```
        self.weights = None
```

```
    def fit(self, X, y):
```

```
        X_b = np.c_[np.ones((len(X), 1)), X]
```

```
        n_features = X_b.shape[1]
```

```

I = np.eye(n_features)
I[0, 0] = 0

self.weights = (
    np.linalg.inv(X_b.T.dot(X_b) + self.alpha * I).dot(X_b.T).dot(y)
)

def predict(self, X):
    X_b = np.c_[np.ones((len(X), 1)), X]
    return X_b.dot(self.weights)

df = pd.read_csv("house_price_data.csv")

X = df[["SquareFeet", "Bedrooms", "Age"]].values
y = df["Price"].values

X_mean = np.mean(X, axis=0)
X_std = np.std(X, axis=0)
X_scaled = (X - X_mean) / X_std

model = RidgeRegressionScratch(alpha=0.1)
model.fit(X_scaled, y)
predictions = model.predict(X_scaled)

mse = np.mean((y - predictions) ** 2)
r2 = 1 - (np.sum((y - predictions) ** 2) / np.sum((y - np.mean(y)) ** 2))

print(f"Intercept (Bias): ${model.weights[0]:.2f}")
print(f"Coefficients (Weights): {model.weights[1:]}")

```

```
print(f"Mean Squared Error: {mse:.2f}")
```

```
print(f"R2 Score: {r2:.4f}")
```

```
sample_house = np.array([[1900, 3, 7]])
```

```
sample_scaled = (sample_house - X_mean) / X_std
```

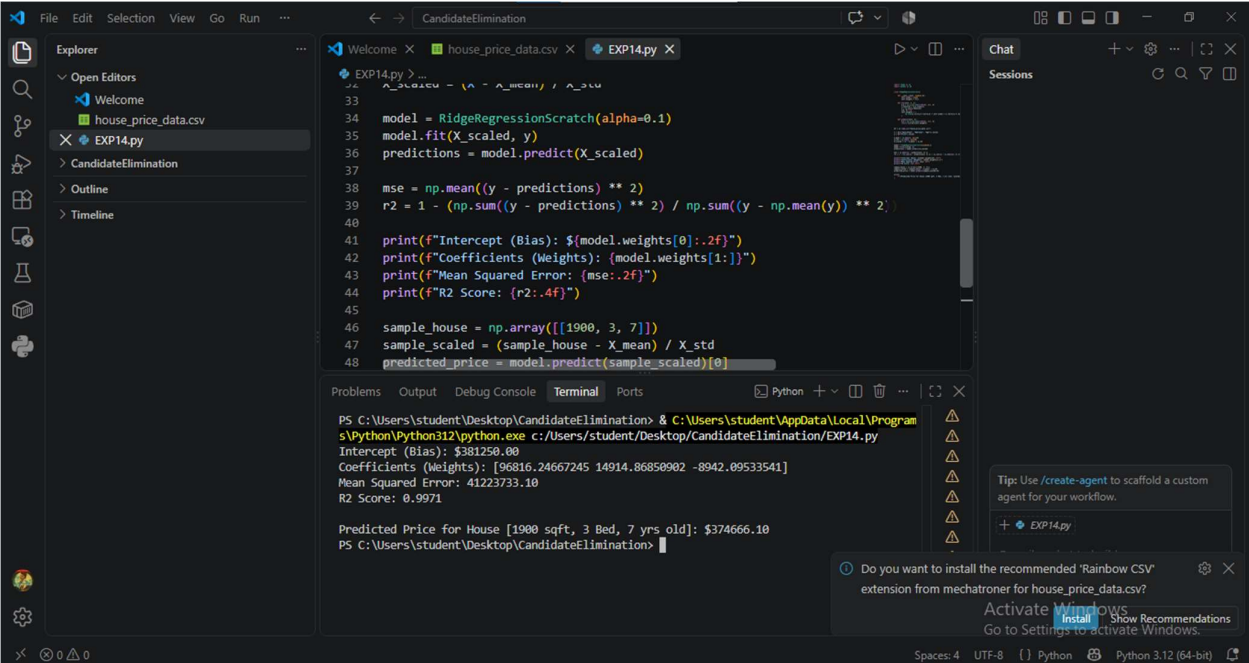
```
predicted_price = model.predict(sample_scaled)[0]
```

```
print(
```

```
    f"\nPredicted Price for House [1900 sqft, 3 Bed, 7 yrs old]: ${predicted_price:.2f}"
```

```
)
```

OUTPUT



The screenshot shows a Visual Studio Code editor with a Python file named `EXP14.py` open. The code implements a Ridge Regression model using `RidgeRegressionScratch` with `alpha=0.1`. It calculates the Mean Squared Error (MSE) and the R2 Score. A sample house data point `[[1900, 3, 7]]` is used to predict the price. The terminal output shows the following results:

```
PS C:\Users\student\Desktop\CandidateElimination> C:\Users\student\AppData\Local\Programs\Python\Python312\python.exe c:/Users/student/Desktop/CandidateElimination/EXP14.py
Intercept (Bias): $381250.00
Coefficients (Weights): [90816.24667245 14914.86850902 -8942.09533541]
Mean Squared Error: 41223733.10
R2 Score: 0.9971

Predicted Price for House [1900 sqft, 3 Bed, 7 yrs old]: $374666.10
PS C:\Users\student\Desktop\CandidateElimination>
```

On the right side of the editor, there is a Chat panel with a tip: "Tip: Use /create-agent to scaffold a custom agent for your workflow." and a button to "EXP14.py". At the bottom right, there is a notification to "Do you want to install the recommended 'Rainbow CSV' extension from mechatroner for house_price_data.csv?" and a button to "Install".